At the Intersection of Graph & Probability Theory

Garrett Kepler

April 10th, 2026

Outline

History of Gerrymandering

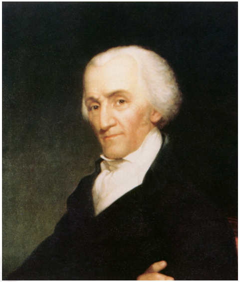

Figure: Eldridge Gerry and his "salamander district" in 1812 Massachusetts: (Click here for full map)

History of Gerrymandering

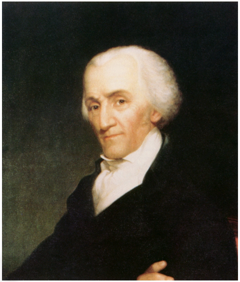

Figure: Eldridge Gerry and his “salamander district” in 1812 Massachusetts: (Click here for full map)

Theorem

Gerry + Salamander = Gerrymander

Proof.

Proof left to audience as an exercise. □

The Maps We See

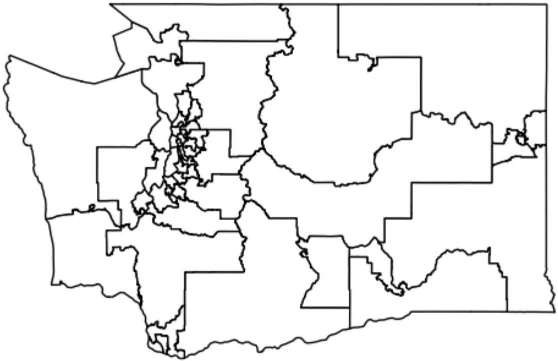

Figure: Washington's 2020 Legislative Districts

The Maps We Deal With

In practice, it is very difficult to deal with maps themselves. So, we simplify them.

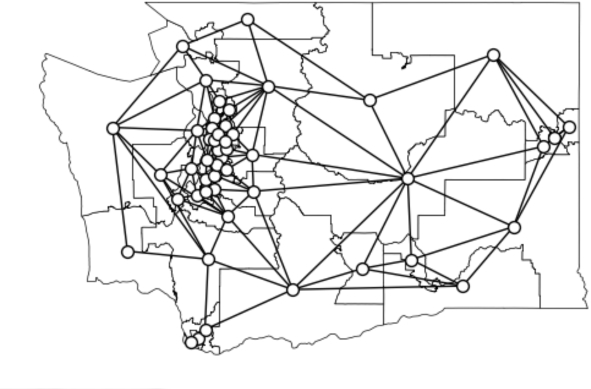

Figure: Washington's 2020 Legislative Districts and corresponding dual graph

The Maps We Deal With

Once we have the dual graphs, we can start thinking about district assignments and metrics of fairness.

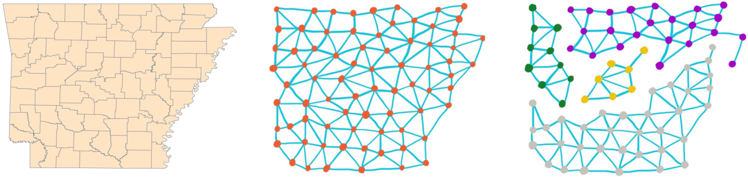

Figure: Example of a dual graph of Arkansas with its district assignment as a partition

Gerrymandering through "Definition"

"*Gerrymandering, defined in the contexts of representative electoral systems, is the political manipulation of electoral district boundaries to advantage a party, group, or socioeconomic class within the constituency.*" - Wikipedia circa 2:53 PM Aprill 6th, 2026 (For Link Click Here)

Question

Okay, but how do we know when someone is trying to advantage a party/group?

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Question

Okay, but how do we know when someone is trying to advantage a party/group?

Answer (*)

We can define any metric we like!

What Might We Use to Determine Gerrymandering?

There are lots of metrics we use on generated maps:

- Mean-Median Score (score far from zero implies district competitiveness is not balanced)
- Polsby-Popper Score (how compact is your district in comparison to the circle)
- Population Balance (are your districts somewhat balanced in population?)
- Majority-Minority Ratio (how well does your districted map respect majority voting power vs. minority voting power)
- And more!

And Now We Compare!

Question

Okay, we have a definition of "fairness" we like, how do we determine if a current map is gerrymandered?

Answer (*)

We can draw all the possible maps and compare!

Womp Womp :(

Answer (*)

We could draw all the possible maps and compare to the map someone wants to enact!

Actually, this highly impractical...

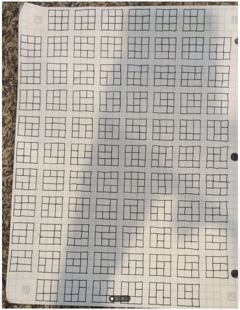
```
In [19]:  start = time.time()
          allperms = list(permutations(partition))
          print('Runtime: '+str(time.time()-start))

          ---------------------------------------------------------------------------
          MemoryError                               Traceback (most recent call last)
          <ipython-input-19-04af173883b2> in <module>
                1 start = time.time()
          ----> 2 allperms = list(permutations(partition))
                3 print('Runtime: '+str(time.time()-start))

          MemoryError:
```

Figure: Memory error for just trying to calculate all 4 partitions of grid from game with 16 nodes (circa 5:12 PM April 6th, 2026)

Perspective



· · · 15 pages later· · ·

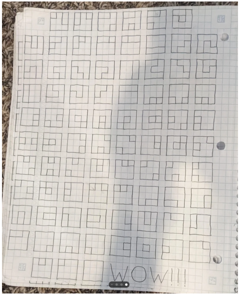


Figure: For perspective, here are the connected partitions of the 3 by 3 grid enumerated by a Reddit user (17 photos total!)

Questions

Question

- What is the *average* of $\{1, 2, 3, 4, 5, 6\}$?
- What is the *expected value* of a dice roll with faces $\{1, 2, 3, 4, 5, 6\}$?

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Intuitively

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Complicatedly

- The average is the sum $\frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$
- The *average of the values we expect to see* is 3.5 ($\mathbb{E}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = 3.5$ where X is the outcome of a dice roll)

Weak Law of Large Numbers (WLLN)

Intuitively

The average should be close to the expected value eventually. That is, if we roll a dice a lot, we should see something close to 3.5 as we continue to roll.

Less Intuitively

The average $\frac{1}{n}\sum_{i=1}^{n} X_i$ will probably converge to $\mathbb{E}[X_i]$ as $n \to \infty$

Complicatedly

If $X_1, X_2, \ldots$ are a sequence of independent identically distributed random variables with $\mathbb{E}[X] < \infty$, then for any $\epsilon > 0$

$$\lim_{n\to\infty} \mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_i]\right| < \epsilon\right] = 1$$

WLLN Through Experience

A little intuition builder:
https://garrett-kepler.github.io/resources/LLN/

A New School of Thought

Figure: (Left) Pavel Nekrasov: firm believer WLLN only applies to independent events. (Right) Andrey Markov: prover that WLLN applies to dependent events too!

WLLN with Dependence

Intuitively

The beautiful result Markov proved is that we actually don't need independent events! With a little assumptions we can confidently say the average will get close to the expected value.

Complicatedly

If $X_1, X_2, \ldots$ is a positive Harris recurrent Markov chain with unique invariant distribution π and f is a real-valued function on the state space, then

$$\hat{\mu_n} = \frac{1}{n}\sum_{i=1}^{n} f(X_i) \xrightarrow{a.s.} \mu = \mathbb{E}_\pi f(X) = \int f(x)\pi(dx)$$

The Recombination Approach

What we do in practice is try to utilize Markov's version of LLN:

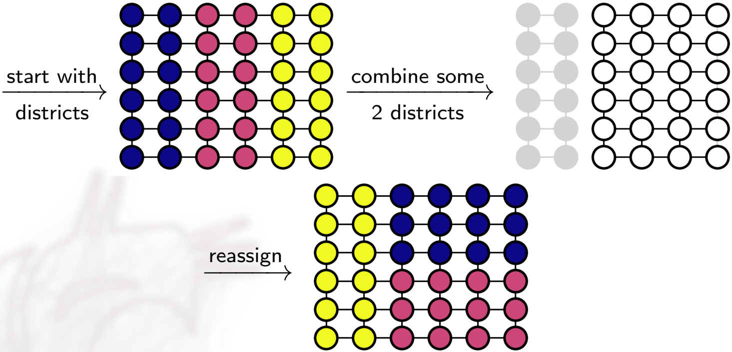
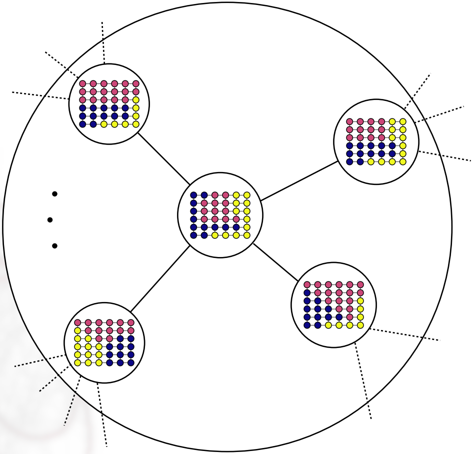


Figure: An example of *dependent events* we're creating

Metagraph

Now, what we get to do is move around the space of districted maps. Moreover, whatever metric we apply will converge to the mean by Markov's LLN!



LLN for Maps

Intuitively

If we recombine and split districts cleverly, we can get a *good* set of maps that we can compare to. That way, we can detect gerrymandering when it occurs!

Less Intuitively

Each movement in the districted maps is a dependent yet random sample such that Markov's version of the Law of Large Numbers applies to any metric we use to detect gerrymandering in maps.

Complicatedly

Ideally, we will define an aperiodic Markov chain with one recurrent class such that the Ergodic theorem applies and we have a unique stationary distribution of districted maps. Then, we can get representative samples from the stationary distribution and can compare.

Ensemble Analysis

This idea is what we call ensemble analysis: Draw a *lot* of *relevant* maps and compare them to proposed maps.

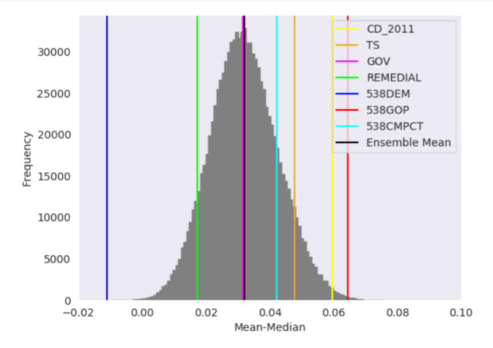


Figure: Comparison of scores from recombination-based maps (grey) with scores of politicians' proposed maps (vertical lines)

Spectral Graph Theory Basics

Theorem (Fiedler Partitioning)

Let G be a connected graph and let x_2 be the eigenvector of the second smallest eigenvalue λ_2 of L. Then, the subgraphs induced by the vertex sets $V_{\geq 0} = \{i : x(i) \geq 0\}$ and $V_{-} = V \setminus V_{\geq 0}$ are connected [2].

Intuitively

The subgraph on the non-negative nodes is connected. Likewise with the negative nodes. We can partition G in to two connected components using the entries of the second eigenvector.

Fiedler Himself

The eigenvector x_2 we colloquially call the "Fiedler vector" after graph theorist Miroslav Fiedler who pioneered this work.

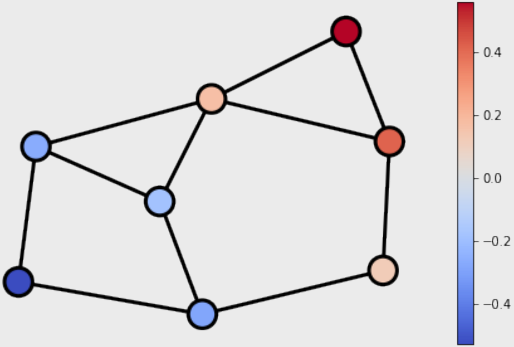
Figure: Miroslav Fiedler (1926-2015)

Fiedler Partitioning

Example

An instance of the Erdös-Renyi graph partitioned using its Fiedler vector

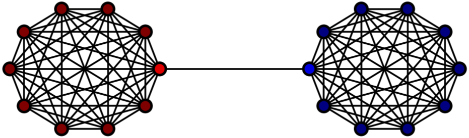
$$x_2 \approx [-0.29, 0.42, 0.56, -0.26, -0.53, -0.19, 0.17, 0.12]^T$$



Fiedler Partitioning

Example

The barbell graph partitioned using its Fiedler vector $x_2 \approx [0.23, 0.23, \dots, 0.23, 0.19, -0.19, -0.23, -0.23, \dots, -0.23]^T$



Fiedler Partitioning

talks/wsu593/photos/starwa

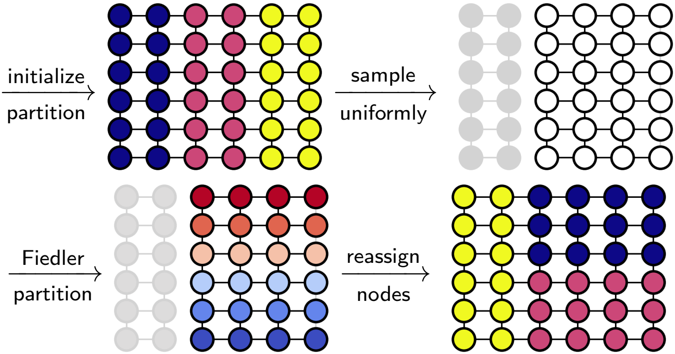
talks/wsu593/photos/jmmpalouse

Figure: (Left) Social network of on-screen conversations in Episode IV of Star Wars. (Right) Pipeline network of the Palouse.

The SpecReCom Algorithm

What we can do is recombine districts and partition using the Fiedler vector:

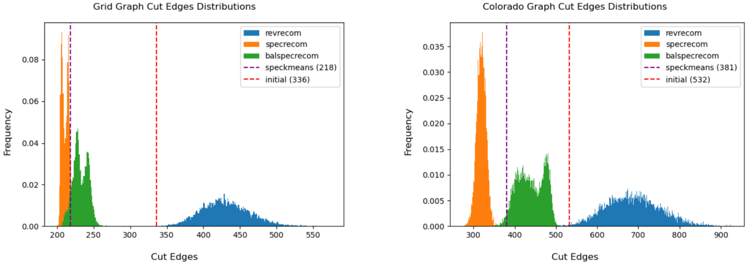
Example



Compactness

Example

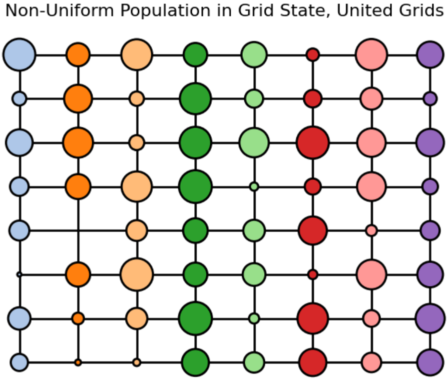
From the original paper [1], Davies et al ran 10,000 chains of SpecReCom and other recombination methods for 400 steps each as a simulation study. Note the dramatic decrease in cut edge counts.



SnailReCom

In collaboration with Phousawanh Peaungvongpakdy: taking in to account the distribution of node attributes. For example, perhaps we care about population distribution in districts.

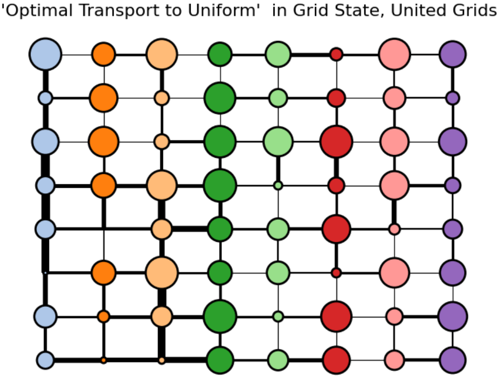
Example



SnailReCom

Embed information about how much work it would take for this network to be "uniform" with respect to population. Randomize according to edge structure.

Example



“Fairness”

Question

So, with all these metrics, when can we say a map is fair?

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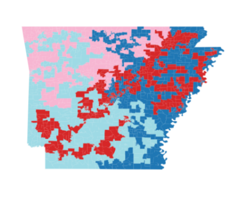
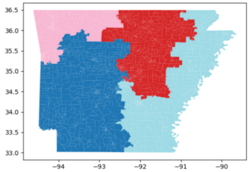
While a map may not pass the "eyeball" test, it may be the best option.





“Fairness”

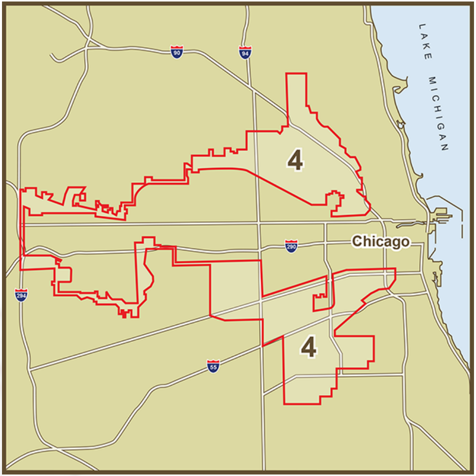
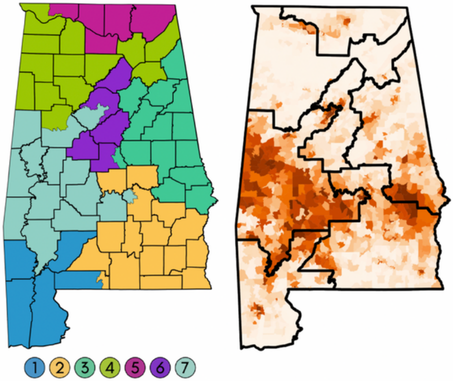


Figure: For example, the famous “Horseshoe District” from Chicago. Making this district score well in compactness would make it score poorly in majority-minority ratio.

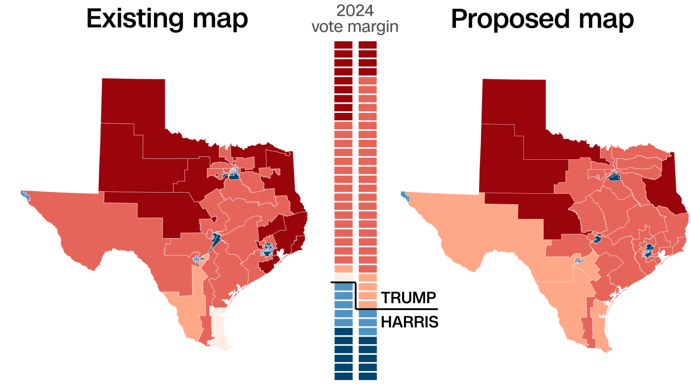
These Questions Still Arise

In 2023, the Supreme Court ruled that Alabama had racially gerrymandered.



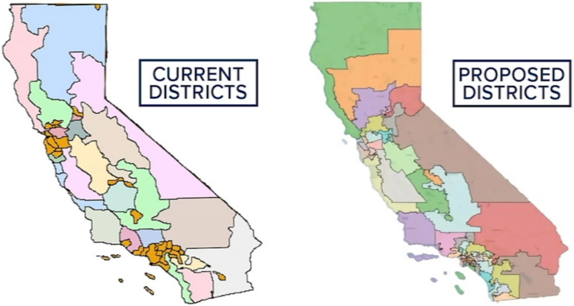
These Questions Still Arise

In 2025, the Supreme Court ruled that Texas had <u>not</u> racially gerrymandered.



These Questions Still Arise

In 2025, California *voted* that they redraw their districted map.



These Questions Still Arise (and There May Be No Answer)

We are not be able to make the perfect map, but we can try and detect bad maps and do better.

Resources

For the curious, here are some redistricting-focused resources:

- "*Political Geometry*" – Textbook on mathematical redistricting methods https://data-democracy.org/gerrybook.html
- "*Redistricting Data Hub*" – Dataset resource with redistricting-oriented data https://redistrictingdatahub.org/

Try Gerrymandering Yourself

Figure: https://garrett-kepler.github.io/resources/gerry/

References

E. Davies, R. Job, M. Kampbell, H. Kim, and H. Seo.
Spectral partitioning of graphs into compact, connected regions, 2025.

M. Fiedler.
A property of eigenvectors of nonnegative symmetric matrices and its application to graph theory.
Czechoslovak Mathematical Journal, 25(4):619–633, 1975.